

imtiazx.ai

Portfolio Design and Narrative Document

Md Imtiaz Ahmed Mollah | AI Engineer | May 2026

VERSION 2

Supersedes v1.0 | PNG sequence approach abandoned | R3F adopted

- | | | | |
|----|--------------------------------|----|--------------------|
| 01 | Overall Narrative | 02 | Homepage Story Arc |
| 03 | Section-by-Section R3F Spreads | 04 | What I Trade |
| 05 | Navigation System | 06 | Homepage Quick-Nav |
| 07 | Full Content Reference | 08 | R3F Implementation |



Table of Contents

imtiazx.ai Portfolio Design Document v2

01	Overall Narrative What changed and what didn't	UPDATED
02	Homepage Story Arc Section order, scroll logic, R3F canvas layer	UPDATED
03	Section-by-Section R3F Spreads S1 through S8 visual specs and transitions	NEW
04	What I Trade Crypto narrative, placement, and content	UNCHANGED
05	Navigation System Nav bar, pages, routes, Crypto tab	UNCHANGED
06	Homepage Quick-Nav Left-edge floating icon navigator (updated icon list)	UPDATED
07	Full Content Reference Projects, posts, identity cards, metrics	UNCHANGED
08	R3F Implementation Architecture Canvas, components, theming, packages, reduced motion	NEW

00

What Changed From v1

Summary of revisions in this revision

REVISIONS

REMOVED FROM V1

PNG frame sequence approach has been retired in full.

- The fixed-position HTML5 Canvas frame player is removed
- lib/narrative.ts (frame index config) is removed
- components/narrative/NarrativeCanvas.tsx is removed
- public/narrative/s1-s2/frames_clean/ frames are retained on disk only as legacy reference; not loaded by the site
- The scroll-to-frame mapper math ($\text{progress} * (\text{totalFrames} - 1)$) is removed
- No more references to 'frame index', 'cover-fit draw', or frame-skip-on-missing in v2 specifications

INTRODUCED IN V2

React Three Fiber visual narrative replaces the frame sequence.

- Each homepage section has a dedicated 3D visual built in R3F + Three.js
- Single persistent R3F Canvas mounted in app/page.tsx, position fixed, z-index 5, pointer-events none, transparent
- useScroll from drei ScrollControls drives all section animations
- Visuals tell a continuous human-AI story; the same character evolves through S1 to S7
- Per-section components live in components/three/: BrainScene, HandsScene, RacingFigure, HybridHand, WritingHand, ThinkerFigure, CryptoHand
- Theme reactivity through MutationObserver on the html element; brain color re-reads --color-brand on theme change
- Reduced motion: useReducedMotion produces a static representative pose for each scene
- What I Trade visual added as a first-class scene (CryptoHand)

PRESERVED FROM V1

- Overall narrative thesis: a human who engineers AI
- Three-act structure: origin and capability, breadth and perspective, values and worldview
- Tagline: I engineer AI systems from prototype to production
- Target audience: senior engineers and hiring managers at product companies
- All content references (projects, posts, identity cards, metrics)
- Navigation structure (Lab, Signal, Stack, About, Earth)
- Homepage Quick-Nav floating-icon feature
- What I Trade content specification
- All hard rules from CLAUDE.md

01

Overall Narrative

The story imtiazx.ai tells and why it matters

UPDATED

CORE IDENTITY

The story of a human who engineers AI.

imtiazx.ai is not a resume rendered as a website. It is a cinematic narrative about a person who sits at the intersection of three disciplines that rarely overlap: rigorous data science, production generative AI engineering, and responsible AI governance. The site tells this story in sequence, using scroll-driven 3D visuals, curated content, and intentional design to communicate to its primary audience: senior engineers and hiring managers at product companies who have seen a thousand portfolios and remember none of them.

THE THREE-ACT STRUCTURE

Act 1: Origin and Capability

Hero and Projects establish who Imtiaz is and what he builds. The R3F brain pulses in the hero, then a human and a robot hand reach in to connect with it as Projects begins. Six radial lines stream from the brain to six project cards. The metaphor is deliberate: Imtiaz bridges human intuition and machine intelligence.

Act 2: Breadth and Perspective

Hackathons, Identity, Writing, and Perspectives establish range. A running figure with an AI head signals competitive instinct. A reaching hybrid hand signals multidisciplinary capability. A writing hybrid hand signals public articulation. A seated, faceless thinker signals depth.

Act 3: Values and Worldview

What I Trade and Earth close the narrative. What I Trade reveals a dimension most AI engineers lack: market intuition, ecosystem literacy, and risk management applied to real capital. Earth grounds everything in a larger worldview: technology in service of something beyond technology.

The Tagline

"I engineer AI systems from prototype to production."
This line lives in the hero and footer. It is precise, confident, and free of buzzwords. It anchors every section.

TARGET AUDIENCE

PRIMARY

Senior engineers at product companies

TONE REGISTER

Premium, engineered, intentional

SECONDARY

Hiring managers and technical recruiters

WHAT THEY FEEL

This person is serious and rare

TERTIARY

Fintech and crypto-native companies

WHAT THEY DO

Save the URL and share it

02

Homepage Story Arc

Section order, scroll logic, and the R3F canvas layer

UPDATED

HOMEPAGE SECTION ORDER (V2)

The homepage is a single continuous scroll. Each section is a chapter, each chapter has a dedicated R3F visual, and the same character evolves across the entire arc. The visual story and the content story run in parallel.

Section	Content	R3F Visual
Hero	Three identity lines. Philosophy. Two Brain	BrainScene: glowing wireframe brain, slow pulse, sparkles, bloom
What I Build (Projects)	6 project cards with status, tags, and Brain	BrainScene + HandsScene: 6 radial lines from brain to cards, human + robot hands reach
What I Compete In (Hackathons)	Flow 3rd place. Rogii Wellbore a Brain	BrainFigure: humanoid body, geometric AI head with glowing visor, motion blur
What I Bring (Identity)	6 identity roles with descriptions.	HybridHand: skin + circuit hand reaches up, drei Html role labels float above
What I Write (Writing)	5 Hashnode articles with categories. Writing	Hand: hybrid hand writes on paper, chart lines emerge into 3D
What I Think (Perspectives)	5 perspective cards on AI and engineering	ThinkerFigure: seated human, face hidden by angle and light, breathing
What I Trade (Crypto)	Crypto analysis since 2018.	CryptoHand: hybrid hand holds BTC, ETH, SOL coins; holographic candlestick chart
Earth	Worldview. Contact. Grounding.	Existing GSAP Tux + background WebM loop (no R3F)

SCROLL-TO-VISUAL LOGIC

A single persistent R3F Canvas covers the full viewport at z-index 5 with pointer-events none. drei ScrollControls provides scroll.offset (0 to 1 across all pages). Each scene component reads its own [enter, exit] offset window from a shared narrative config and lerps its state. Bloom and tone mapping are shared across all scenes via a single EffectComposer.

```
progress = clamp((scroll.offset - sceneEnter) / (sceneExit - sceneEnter), 0, 1)
```

Each scene is mounted/visible only while progress is in [0, 1] or within a small lead-in/lead-out band, so the GPU is not rendering inactive scenes.

THEME REACTIVITY

On mount, each scene reads --color-brand from getComputedStyle(document.documentElement) and stores it as a uniform. A MutationObserver watches the html element's class attribute and re-reads the variable on theme change, then updates the uniform without remounting.

03

Section-by-Section R3F Spreads

S1 through S8 content, visual, and transitions

NEW

Each of the eight homepage sections gets its own dedicated spread. Left column holds the content specification. Right column holds the visual description, R3F implementation notes, scroll trigger, and transition to the next section.

S1 -- Hero

id: narrative-s1

CONTENT

- Three identity lines in Geist Sans
Lines: "Data Scientist by origin." / "Generative AI Engineer by evolution." / "Forward Deployed Engineer by trajectory."
- Role weight 500 in --color-text-primary, progression word weight 500 in --color-brand at 0.9 opacity, connecting 'by' weight 300 in --color-text-muted
- Philosophy line: "I work at the intersection of AI research prototypes, production systems, and human chaos."
- Two CTAs: View the Lab (primary brand) and Read Signal (ghost)
- Section background: theme-adaptive dark panel

LAYOUT

- Two columns: left holds identity text, right hosts the R3F brain
- Right column is empty DOM, the persistent R3F Canvas overlays it
- Subtle dark left-side scrim for text readability when canvas is opaque
- min-height: 100vh, scroll exits into S2

VISUAL (R3F BRAIN SCENE)

- Low-poly wireframe brain, IcosahedronGeometry subdivided 3x
- Material: MeshStandardMaterial with emissive set to --color-brand, emissiveIntensity 1.2 baseline
- Slow continuous Y-axis rotation, 0.18 rad/s
- Heartbeat pulse: sharp attack 80ms, slow decay 1100ms, period 1.2s. Drives emissiveIntensity 1.0 to 2.4
- Sparkles from drei float around the brain, count 80, noise 0.6, speed 0.4
- Bloom postprocessing: luminanceThreshold 0.2, intensity 1.6, radius 0.85
- Brain color reads getComputedStyle(html, --color-brand), updates on MutationObserver(class)
- Canvas background transparent so the page background shows through

TRANSITION TO S2

- Drives the last 30% of S1 scroll range
- position.x lerps from 1.8 to 0.0
- scale lerps from 1.0 to 0.7
- Human hand slides in from screen-left, robot hand from screen-right
- Orange energy arcs (drei Line, dashed, animated) connect both fingertips to brain surface

S2 -- What I Build (Projects)

id: narrative-s2

CONTENT

- Section heading: What I Build
- 6 project cards from lib/projects.ts
- Card format: name, subtitle, status badge, description, tag chips, links to repo and details
- Status badge types: Production (teal), Development (brand), Ideation (amber)
- Filter by status and origin lives on /lab page only, not on the homepage

LAYOUT

- Grid: 1 col mobile, 2 col tablet, 3 col desktop
- Cards: position relative, z-index above the R3F canvas
- Card hover: subtle lift, brand accent strengthens on left border
- Section vertical rhythm: clamp() padding aligned with rest of homepage

VISUAL (R3F PROJECT CONSTELLATION)

- Brain reaches center when S2 top is reached (final transition state from S1)
- 6 glowing radial lines (drei Line) at 60-degree intervals point outward from brain center toward project cards
- Lines animate in sequentially when section scroll progress > 0.3
- Traveling dots (GSAP MotionPathPlugin) ride the lines from brain to card endpoints, 1 dot per line, 220ms offset
- Human hand reaches in from screen-left, robot hand from screen-right, both connected to brain via orange energy arcs
- Hands remain visible throughout S2
- Lines use --color-brand at 0.6 opacity, dots at full opacity

TRANSITION TO S3

- Hands retract toward edges and fade
- Brain shrinks, rises off-screen vertically
- Racing figure enters from screen-left at S3 top

S3 -- What I Compete In (Hackathons)

id: narrative-s3

CONTENT

- Section heading: What I Compete In
- Active: Rogii Wellbore Intelligence (currently #115 globally)
- Completed: LangFlow Global Hackathon (3rd place)
- Card format: competition name, rank or status, description, stack used
- Status badges: Active (pulsing brand dot) and Completed (teal)
- Source of data: lib/hackathons.ts

LAYOUT

- Two columns desktop, one column mobile
- Section transitions cinematically: figure runs behind cards
- Dark theme-adaptive surface, cards above R3F layer

VISUAL (R3F RACING FIGURE)

- Humanoid in a running pose: human body, geometric AI robot head
- Body: Mixamo GLB with CC0 running animation, skinned mesh
- Head: custom IcosahedronGeometry replaces the original Mixamo head bone, glowing brand-orange visor strip
- Robot head emissive intensity 1.4 on visor only
- Limb motion blur via R3F EffectComposer MotionBlur pass
- Background plane: dark racing track texture with brand-orange lane lines, scrolls horizontally to suggest motion
- Run animation speed = base 1.0 plus scroll velocity contribution, clamped [0.7, 2.4]

TRANSITION TO S4

- Figure decelerates, plants stride, raises one arm upward
- Arm geometry morphs into the hybrid hand of S4
- Camera pushes in toward the rising hand

S4 -- What I Bring (Identity)

id: narrative-s4

CONTENT (6 IDENTITY ROLES)

- GenAI Engineer: RAG systems, agentic pipelines, LLM APIs, production evals
- Responsible AI: bias auditing, safety evals, red-teaming, governance at scale
- Data Scientist: statistical modeling, ML pipelines, experiment design
- System Designer: API architecture, latency and cost tradeoffs, infra patterns
- AI Consultant: translating business problems to deployed AI end to end
- Technical Writer: deep-dive articles, open-source docs, public learning

LAYOUT

- 6 role cards arranged in a 2x3 grid (3x2 on wide viewports)
- Card surfaces above R3F canvas, drei Html labels rise from below
- Labels use --color-text-primary and --font-sans

VISUAL (R3F HYBRID HAND)

- Close-up hand reaching up from the bottom of the viewport
- Palm: human skin texture, fingers etched with circuit lines and metallic joint segments
- Built from a CC0 hand GLB plus a custom mix shader that blends skin and circuit textures via a uniform mix factor (0.5 baseline, subtle 8s ease in and out)
- Dramatic upward lighting: warm rim light from below, cool fill from above-left
- drei Html labels (6 role names) positioned above the open palm, occlude=blend so they read on the dark background
- Labels rise upward staggered 120ms apart as scroll advances

TRANSITION TO S5

- Hand tilts, fingers curl around a pen primitive
- Paper plane appears below, lit faintly orange at the edges

S5 -- What I Write (Writing)

id: narrative-s5

CONTENT (5 POSTS)

- Scaling GenAI from POC to enterprise production (Technical)
- Deception and autonomy in LLMs, a DeepSeek R1 study (Responsible AI)
- Pricing agentic AI, a value-based framework (Strategy)
- ARC-AGI 2024, what 50% GPT-4o accuracy means (Research)
- Chunking strategies that affect RAG quality (Technical)
- All hosted on Hashnode at blog.imtiax.dev
- Card hover: post title scales up, category chip glows in brand
- Read all on Signal CTA links to /signal page

VISUAL (R3F WRITING HAND)

- Same hybrid hand from S4 carried forward for continuity
- Hand now grips a pen, writes on a paper plane positioned below it
- Paper glows faintly at the edges in brand orange
- Abstract line-chart geometry emerges from the writing as scroll advances, rendered with drei Line components in 3D space
- Chart represents analytical thinking made visible, not data
- Micro-variation: hand alternates between skin and circuit appearance on a slow oscillation (10s) so the hybrid nature reads
- Lighting: warm focal lamp from upper-left, deep shadow elsewhere

TRANSITION TO S6

- Pen and paper dissolve into faint particle wisps
- Camera pulls back, hand exits the frame
- Seated figure resolves from low ambient light

S6 -- What I Think (Perspectives)

id: narrative-s6

CONTENT (5 PERSPECTIVE CARDS)

- Card format: bold claim, 2 to 3 sentence elaboration
- Topics: AI autonomy, responsible AI, engineering philosophy, the future of human-AI collaboration, and what gets lost in the race for capability
- Source: lib/opinions.ts (5 cards)
- Section background: theme-adaptive --color-surface-alt; perspective card surfaces stay fixed dark for editorial tone (per CLAUDE.md hard rule, documented exception)
- Cards use light text on dark card surfaces

VISUAL (R3F THINKER FIGURE)

- Human figure seated, leaning forward slightly, viewed from behind at a 45-degree angle
- Face fully hidden by camera angle and very low ambient lighting
- Single rim light from upper-left catches shoulder and side of head
- No circuits, no robot parts, no brand glow - the most human moment in the narrative
- Quiet, contemplative, private mood
- Rendering: PBR materials, drei AccumulativeShadows for grounding, area lighting, ambient occlusion
- Subtle 3-second breathing cycle on chest scale (0.99 to 1.01)
- No other motion - stillness is the point

TRANSITION TO S7

- Figure dissolves into faint sparkles
- Coin geometries materialize in darkness ahead of the next visual

S7 -- What I Trade (Crypto)

id: narrative-s7

CONTENT

- Section heading: What I Trade
- Subheading: "On-chain since 2018. Six years of ecosystem research, risk analysis, and portfolio management across market cycles."
- Six stat cards: Active since 2018, Twitter following 2000+, 2 full market cycles, 15+ testnets tested, On-chain plus fundamentals, Coverage L1 / L2 / DeFi / RWA

THREE ANALYSIS PILLARS

- Ecosystem Research: protocol whitepapers, tokenomics, VC backing, developer activity, on-chain metrics
- Portfolio Management: multi-cycle experience, bear and bull playbooks, risk-adjusted position sizing
- Community: 2000+ Twitter followers, original analysis only, no signals, public track record

FEATURED PROTOCOLS

- Polymarket: prediction markets, testnet UX feedback during early access
- Aethir: decentralized GPU compute, selected as early tester, feedback on node performance and staking mechanics

DATA SOURCE

All content lives in **lib/crypto.ts** (new file).

VISUAL (R3F CRYPTO HAND)

- Same hybrid hand aesthetic as S4 and S5 (character continuity)
- Hand holds three floating coins above an open palm
- Bitcoin: gold-tinted disc with B symbol
- Ethereum: silver disc with diamond shape
- Solana: purple-teal gradient disc with S symbol
- Coins float and rotate slowly, staggered phase
- Holographic candlestick chart rises above the coins in brand-orange wireframe (drei Line + Sparkles)
- Chart animates: candles form and resolve, moving average traces through them
- Background: pure dark with faint blockchain node-and-edge network pattern at 6% opacity

SCROLL BEHAVIOR + TRANSITION TO S8

- Coins float in one at a time as scroll enters S7: BTC first, then ETH, then SOL
- Chart builds progressively across S7 scroll range
- Transition: coins dissolve, hand lowers and exits frame
- Greenery and atmospheric haze fade in to introduce Earth

S8 -- Earth

id: earth

CONTENT (EXISTING)

- Section id=earth, scroll-target for the Earth nav button (homepage only)
- Worldview statement from lib/person.ts
- Contact CTA: email link
- Existing GSAP Tux animation scene preserved entirely
- Fixed dark background #0C1A0E (CLAUDE.md documented exception)

NO R3F IN THIS SECTION

- Earth is the resting end of the narrative; existing implementation stays
- Persistent R3F Canvas is hidden (display none) once S7 has exited the viewport
- GSAP Tux scene preserved untouched per CLAUDE.md hard rule

VISUAL (BACKGROUND VIDEO LAYER)

- WebM video loop layered behind the existing Tux canvas at a lower z-index
- Content: lush terrain, distant iceberg, atmospheric haze, golden cinematic light
- Muted, autoplay, playsInline, loop, poster frame shown until loaded
- Video covers full viewport, object-fit cover, slight blur (3px) so Tux reads clearly above
- Pause/resume tied to IntersectionObserver to save bandwidth when off-screen

THEME CONTINUITY

- Earth section preserves its fixed dark background as before
- Footer CTA renders beneath (existing implementation)
- Brand orange tones return in the contact CTA

04

What I Trade

Full specification for the crypto and fintech narrative section

UNCHANGED

RATIONALE

Most AI engineers have no financial market experience. Intiaz has been actively analyzing and trading crypto since 2018 - through two full bull-bear cycles. This is not a hobby footnote. It is a demonstration of the same skills that make a great AI engineer: pattern recognition under uncertainty, risk modeling, ecosystem analysis, and the discipline to manage positions across volatile conditions. For fintech employers, this section is the differentiator.

FULL CONTENT SPECIFICATION

"What I Trade"
"On-chain since 2018. Six years of ecosystem research, risk analysis, and portfolio management across every market cycle."

SIX STAT CARDS

ACTIVE SINCE	TWITTER FOLLOWING	MARKET CYCLES
2018	2,000+	2 full cycles
PROTOCOLS TESTED	ANALYSIS FORMAT	COVERAGE
15+ testnets	On-chain + fundamentals	L1s, L2s, DeFi, RWA

WHAT I ANALYZE -- THREE COLUMNS

Ecosystem Research Protocol whitepapers, tokenomics models, team backgrounds, VC backing, on-chain metrics, developer activity.	Portfolio Construction Core (BTC and ETH), satellite (L1 challengers, L2s), speculative (early-stage protocols with testnet activity).	Market Cycle Management Position sizing by conviction tier, stablecoin allocation in bear markets, DCA strategies across volatility regimes.
Red Flag Detection Unsustainable APY structures, anonymous teams with no track record, token unlock schedules designed to dump on retail, liquidity pool manipulation patterns.	Community Analysis Twitter sentiment, Discord activity quality, governance participation rates as signal for protocol health.	On-chain Data Wallet tracking, exchange flow analysis, large holder behavior, smart money wallet following.

Protocols I have shaped

Polymarket

Prediction
Markets

Prediction market protocol

Reached out during testnet phase. Provided UX feedback on market creation flow and resolution mechanics. Active user since early access.

Aethir

DePIN /
Compute

Decentralized GPU compute network

Selected as early tester during testnet. Evaluated node performance, staking mechanics, and compute marketplace UX. Provided structured feedback.

TWITTER PRESENCE

2,000+ followers built through consistent, original analysis posts. Content includes protocol deep-dives, market structure commentary, and testnet experience reports. No copy-trading signals - all original research. Twitter handle lives in lib/person.ts and is surfaced in the What I Trade section and the About page.

PLACEMENT DECISION

- Section ordering in v2: inserted between Perspectives (S6) and Earth (S8) as S7
- Rationale: closes Act 3 (values and worldview) before the Earth grounding moment
- Fintech audience sees this near the end of the narrative arc, after the AI story is established
- Does not interrupt the AI narrative arc

05

Navigation System

Nav bar, all pages, routes, updated structure

UNCHANGED

NAV BAR

The top navigation bar structure is unchanged. What I Trade does NOT get its own nav item; the nav stays intentionally minimal. The crypto section surfaces through three access points: the homepage section, a Crypto tab on the Signal page, and a bio mention on the About page.

CURRENT NAV STRUCTURE

Nav Item	Route	Type	Content
Lab	/lab	Page	Projects + Hackathons + Open Source tabs
Signal	/signal	Page	Articles + Perspectives + Crypto tabs
Stack	/stack	Page	Tech categories with proficiency bars
About	/about	Page	Bio + social + timeline + Right Now
Earth	#earth	Scroll button	Homepage only, smooth scroll to Earth section

PAGE CONTENT DETAIL

/lab

- Tab 1 - Projects: 6 projects, status filter (All / Production / Development / Ideation), origin filter (Personal / Enterprise-inspired)
- Tab 2 - Hackathons: active and completed, card with rank badge
- Tab 3 - Open Source: placeholder for future contributions

/signal

- Tab 1 - Articles: 5 Hashnode posts with category filter (All / Technical / Responsible AI / Strategy / Research)
- Tab 2 - Perspectives: 5 opinion cards from lib/opinions.ts
- Tab 3 - Crypto: crypto analysis posts and protocol reviews, link to Twitter

/stack

- 5 tech categories with proficiency indicators and usage notes
- Categories: LLM APIs, RAG and Retrieval, Agent Frameworks, Backend, Frontend and Viz
- Each item: name, proficiency bar, note on how it is used in real work

/about

- Bio paragraph: background, Accenture responsible AI work, crypto since 2018 mention
- Social links: GitHub, LinkedIn, Hashnode, Twitter (all from lib/person.ts)
- Terminal CodeBlock: career timeline aesthetic
- Timeline: key career milestones from data science to GenAI engineering
- Right Now cards: current focus, active hackathon, latest post, current reads

06

Homepage Quick-Nav Feature

Left-edge floating icon navigator with updated icon list

UPDATED

FEATURE OVERVIEW

A vertical strip of icons anchored to the left-center of the viewport. Auto-hidden by default - invisible until the user moves their cursor into a 48px trigger zone on the left edge. On hover-in, the icons slide in from the left with a spring animation. Each icon represents one homepage section and clicking it smooth-scrolls to that section. This is distinct from the top navigation bar, which navigates between pages.

BEHAVIOUR SPECIFICATION

Trigger and reveal

- Trigger zone: 48px wide strip on the left viewport edge, full height
- On mouse enter trigger zone: icons slide in from left, opacity 0 to 1
- Spring animation via Framer Motion: stiffness 300, damping 28
- Each icon staggers 40ms after previous
- On mouse leave the full panel (icons + trigger zone): slide back out
- Delay on hide: 400ms so accidental exits do not flicker

Active state

- Icon for the currently visible section is highlighted
- Highlight: --color-brand fill, slight scale up (1.15x)
- Determined by IntersectionObserver on each section
- Smooth transitions between active states as user scrolls

ICONS AND SECTIONS (LUCIDE REACT)

- Hero: Cpu icon (AI brain)
- What I Build: Layers icon (stacked projects)
- What I Compete: Trophy icon (hackathons)
- What I Bring: Briefcase icon (identity)
- What I Write: PenLine icon (signal / writing)
- What I Think: MessageSquare icon (perspectives)
- What I Trade: TrendingUp icon (crypto / fintech) [NEW]
- Earth: Globe icon (earth section)

Tooltip

- Section name appears as a tooltip to the right of each icon on hover
- Tooltip uses --font-mono, 10px, --color-text-primary
- Tooltip fades in 80ms after icon hover

TECHNICAL IMPLEMENTATION

Component: components/ui/HomepageQuickNav.tsx

- position fixed, left 0, top 50%, transform translateY(-50%), z-index 20
- Trigger zone: a 48x100vh invisible div at left 0
- Icon panel: flex column, gap 16px, padding 12px 10px
- Panel background: rgba(0,0,0,0.6) with 8px blur backdrop-filter
- Border: 1px solid rgba(255,255,255,0.08) on right and top/bottom edges
- Border radius: 0 8px 8px 0 (left edge flush with viewport)
- Each icon: 28px x 28px, Lucide React, strokeWidth 1.5
- Default color: --color-text-muted. Active: --color-brand
- On click: window.scrollTo with behavior smooth to section offsetTop
- Hidden on mobile (viewport width below 768px)
- Respects prefers-reduced-motion: no slide animation, instant show/hide

07

Full Content Reference

All structured content across lib/ files

UNCHANGED

PROJECTS (LIB/PROJECTS.TS)

RAGScope

DEVELOPMENT

Retrieval evaluation harness | RAG, RAGAS, pgvector, FastAPI, LangSmith

Benchmarks 5 RAG strategies side by side using RAGAS metrics. Live dashboard shows faithfulness, latency, and context precision.

DocuAgent

IDEATION

Multi-agent document intelligence | LangGraph, Claude API, Redis, WebSockets

Supervisor agent decomposes uploaded documents into tasks and routes to specialist sub-agents for clause extraction and risk annotation.

SafeScore

IDEATION

Responsible AI evaluation toolkit | Responsible AI, Red-teaming, FastAPI, React

Red-teaming for deployed GenAI applications. Tests attack surface from system prompts, RAG pipelines, and agentic tool configurations.

LLM Observatory

IDEATION

Production monitoring for LLM applications | Python SDK, FastAPI, TimescaleDB, Docker

Drop-in SDK and self-hostable dashboard capturing token usage, cost, latency p50 / p95, and output quality over time.

GraphRAG Lite

IDEATION

Knowledge graph enhanced retrieval | Knowledge Graph, NetworkX, Qdrant, Claude API, D3

Combines entity extraction, graph traversal, and vector search. Shows reasoning path behind every retrieved result.

PromptOps

IDEATION

Prompt version control and A/B testing | FastAPI, PostgreSQL, Redis, React

Version control for prompts with built-in A/B testing and Claude-as-judge evaluation. Visual diff viewer.

WRITING POSTS (LIB/POSTS.TS)

Category	Title
Technical	Scaling GenAI from POC to enterprise production
Responsible AI	Deception and autonomy in LLMs - a DeepSeek R1 study
Strategy	Pricing agentic AI - a value-based framework
Research	ARC-AGI 2024 - what 50% GPT-4o accuracy means
Technical	Chunking strategies that affect RAG quality

IDENTITY CARDS (LIB/PERSON.TS)

Role	Description
GenAI Engineer	RAG systems, agentic pipelines, LLM APIs, production evals
Responsible AI	Bias auditing, safety evals, red-teaming, governance at scale
Data Scientist	Statistical modeling, ML pipelines, experiment design
System Designer	API architecture, latency and cost tradeoffs, infra patterns
AI Consultant	Translating business problems into deployed AI end to end
Technical Writer	Deep-dive articles, open-source docs, public learning

METRICS (LIB/PERSON.TS - ABOUT PAGE)

YEARS SHIPPING PRODUCTION AI 4+	ENTERPRISE GENAI STUDIES 25+	EFFORT REDUCTION DELIVERED 75%
LANGFLOW GLOBAL HACKATHON 3rd place	ROGII WELLBORE (LIVE) #115 global	CRYPTO ON-CHAIN SINCE 2018

08

R3F Implementation Architecture

React Three Fiber stack, components, theming, reduced motion

NEW

PACKAGES TO INSTALL

```
npm install @react-three/fiber @react-three/drei @react-three/postprocessing three @types/three
```

PERSISTENT CANVAS

A single R3F Canvas is mounted in app/page.tsx as the very first child of the root page div. The Canvas wraps drei ScrollControls so all scene components share a single scroll source. The Canvas is styled position fixed, top 0, left 0, width 100vw, height 100vh, z-index 5, pointer-events none, background transparent.

COMPONENT INVENTORY (COMPONENTS/THREE/)

Component	Section	Role
BrainScene.tsx	S1, S2 (carryover)	Wireframe brain, pulse, sparkles, bloom
HandsScene.tsx	S2	Human + robot hands, energy arcs, radial lines to project cards
RacingFigure.tsx	S3	Mixamo body + custom AI head, motion-blurred running stride
HybridHand.tsx	S4	Skin+circuit hand reaching upward, drei Html role labels
WritingHand.tsx	S5	Hybrid hand writing on paper, emerging chart lines
ThinkerFigure.tsx	S6	Seated human, face hidden, contemplative stillness
CryptoHand.tsx	S7	Hybrid hand with BTC/ETH/SOL coins, holographic chart
narrative.ts	lib/	Per-scene enter/exit scroll offsets, asset paths, GLB references

ASSET SOURCES

- Mixamo CC0 humanoid rigs for racing body and hybrid-hand base
- Sketchfab CC0 hand GLB for HybridHand/WritingHand/CryptoHand
- Coins generated procedurally with R3F primitives (TorusGeometry with custom shader), not GLB
- Brain geometry: IcosahedronGeometry subdivided 3x, wireframe material
- All textures live under public/three/textures/ as compressed AVIF
- Total payload target: under 6MB across all scenes combined (Vercel free tier guardrail)

THEME REACTIVITY

Each scene reads --color-brand from getComputedStyle on mount, stores it in a Three.Color uniform, and updates that uniform when a MutationObserver on the html element fires for class attribute changes (theme toggle). No remount required. The same observer is shared by all scenes via a small useBrandColor() hook in components/three/hooks/.

REDUCED MOTION

useReducedMotion (Framer Motion hook) gates every animation. When set, scenes render a single representative pose:

- BrainScene: brain visible, no pulse, no rotation
- HandsScene: full constellation rendered statically, no traveling dots
- RacingFigure: figure in stride pose, no motion blur, no run loop
- HybridHand: hand fully extended, labels visible without rise animation
- WritingHand: hand mid-stroke, chart fully revealed
- ThinkerFigure: figure still, no breathing cycle
- CryptoHand: all three coins visible, chart fully drawn

FALLBACK FOR R3F-INCAPABLE CLIENTS

On WebGL context creation failure, the Canvas onCreated handler swaps to a server-rendered static SVG illustration set keyed by section id. This keeps the narrative legible on low-end devices and screen readers without leaving a blank viewport.

HARD RULES FROM CLAUDE.MD

- Never use em-dashes anywhere in code, content, or comments
- Never use emojis anywhere
- Never hardcode content inside components - all content from lib/
- Never invent new color values - use CSS custom properties from globals.css only
- Run npm run build before considering any task complete
- Test all three themes (light, dark, system) after every UI change
- Test audio toggle in all three states after any audio-related change
- Verify prefers-reduced-motion disables all animation
- Two hardcoded bg exceptions: EarthSection #0C1A0E and PerspectivesSection cards #1C1412

NEXT IMPLEMENTATION TASKS IN ORDER

1	Install R3F packages and scaffold components/three/
2	Build BrainScene with pulse, rotation, sparkles, bloom
3	Mount persistent Canvas in app/page.tsx with ScrollControls

4	Build HandsScene (human + robot hands, radial lines)
5	Wire BrainScene + HandsScene transition (S1 to S2)
6	Build RacingFigure with Mixamo GLB + custom head
7	Build HybridHand with skin+circuit mix shader
8	Build WritingHand variant with emerging chart
9	Build ThinkerFigure (PBR, area lighting, breathing)
10	Build CryptoHand with coins and holographic chart
11	Add Earth background WebM loop behind existing Tux scene
12	Wire useBrandColor() theme observer across all scenes
13	Wire useReducedMotion fallback poses across all scenes
14	Add WebGL failure fallback (static SVG per section)
15	Run npm run build and test all three themes